

Tony Lin

(647) 640-5703 | tonylin305@gmail.com | linkedin.com/in/itstonylin | github.com/itstonylin | itstonylin.ca

EDUCATION

Bachelor of Applied Science, Honours Computer Science Co-op

McMaster University, Summa Cum Laude

Sep 2021 – Apr 2026

Hamilton, Ontario

WORK EXPERIENCE

Lead Software Engineer

Jun 2025 – Present

Backyard Brands | *React Native, TypeScript, Kotlin, Swift, PostgreSQL, PHP, OpenAI, ElevenLabs*

Pickering, Ontario

- Lead a team of 5 developers building bilingual pool-care mobile apps with conversational AI, water testing, and job management, used by 50+ partner retail locations across Canada.
- Developed native iOS and Android integrations for Solos smart glasses, enabling pool-care technicians to access conversational AI hands-free and capture photos for AI-powered image analysis.
- Connected printed Aqua Ace water-test prescriptions to mobile treatment guides via QR codes, adding dose timers, progress tracking, conversational AI support, and retailer analytics.
- Developed a bilingual, location-aware weather alert system for 5,000+ pool owners, delivering forecast-triggered product recommendations and driving an 8% increase in attributed sales.

Teaching Assistant

Sep 2023 – Apr 2026

McMaster University

Hamilton, Ontario

- Supported students and evaluated assignments for cohorts of 180+ across five terms and three courses.
- COMPSCI 2ME3: Introduction to Software Development (Fall 2023, 2024, 2025); SFWRENG 2OP3: Object-Oriented Programming (Fall 2025); COMPSCI 1XC3: CS Practice and Experience: Development Basics (Winter 2025, 2026).

Process Automation Engineer

Jan 2024 – Dec 2024

Stelco | *Python, PL/SQL, Bash, Power Automate, Oracle Analytics Publisher, Power BI, Git*

Nanticoke, Ontario

- Automated steel-batch chemistry deviation detection, reducing metallurgist review time by 95%.
- Automated bulk report uploads and linking with Python and Power Automate, eliminating repetitive GUI work and reducing deployment time from hours to minutes.
- Migrated and deployed 140+ Oracle Analytics Publisher reports across development, testing, and production servers, maintaining full functionality for previewing, scheduling, and automated distribution.
- Added search and quick-action shortcuts to the Lake Erie Steelmaking reporting utility using PL/SQL, reducing report retrieval time by 60%.

Software Engineer

May 2022 – Aug 2022, May 2023 – Aug 2023

Tech Mahindra | *Java, JUnit, Oracle SQL, UML, Git*

Toronto, Ontario

- Built a Java plugin to generate JUnit test scaffolding for Spring Boot APIs, accelerating test development.
- Normalized database schemas in Oracle SQL, eliminating redundant data and reducing storage footprint by 10%.
- Authored technical specifications for a mortgage-processing system, identifying validation gaps and documenting edge cases to expand test coverage.

PROJECTS

ApptVia | *Next.js, TypeScript, PostgreSQL, Supabase, OpenAI, ElevenLabs, Stripe, Twilio, Tailwind* Jun 2026 – Present

- Building an AI-powered scheduling platform for service businesses with online booking, configurable availability, customer management, payments, notifications, and field-service routing.
- Engineering a multi-channel scheduling agent that executes booking and staff-scheduling requests under configurable rules and approval controls.

Coping-Gacha | *React Native, Expo, TypeScript, Supabase, Claude API, PostgreSQL, Git*

Jan 2026 – Apr 2026

- Developed a therapeutic app with 10 AI personalities offering distinct perspectives on personal challenges.
- Engineered cross-conversation memory with Claude Haiku summaries to preserve context across AI personalities.
- Built a research analytics dashboard tracking user actions and engagement across therapeutic modalities.
- Co-authored a Graphics Interface 2026 extended abstract and research poster on pluralistic AI coping support.

Peerio | *React Native, Expo, Supabase, Python, Railway, Git*

Sep 2025 – Apr 2026

- Built a mobile app with a 5-person team enabling two-way matching between people seeking project experience and project creators looking for collaborators through Tinder-style swiping.
- Developed a Python-based hybrid matching engine using sentence transformers for semantic similarity, skill overlap scoring, and Elo ranking, deployed on Railway.

TECHNICAL SKILLS

Languages: TypeScript, Python, JavaScript, Java, Kotlin, Swift, SQL, PHP, C/C++, Bash, HTML/CSS

Frameworks & Libraries: React Native, React, Next.js, Node.js, Expo, Tailwind, Vue.js, JUnit

Databases & Platforms: PostgreSQL, Supabase, MySQL, Vercel, Railway

Tools & Integrations: Git, Docker, OpenAI, Claude, ElevenLabs, Stripe, Twilio